

Build your own agent: setup guide

What you are building

You are about to set up a small autonomous AI agent that wakes itself up every day at 9 AM Eastern Time, decides what to think about, writes a short entry in a public diary, and sends you a private message on Telegram with whatever it wants you to know. It runs entirely on free services. Your running cost is \$0 per month, forever.

What it will cost

- One-time setup: 30 to 45 minutes of your time
- Daily running: \$0 forever (free tiers only)
- No credit card required for any account in this guide

Accounts you need (all free, no card required)

You need four accounts. Set them up in this order, because each later step depends on the earlier ones.

1. **GitHub account:** <https://github.com/signup>. Where the agent's code, state, and daily logs live. The cron that wakes the agent runs on GitHub's servers.
2. **OpenRouter account:** <https://openrouter.ai/sign-up>. Gives the agent its language model brain. Free tier includes models like Llama 3.3 70B, Qwen 80B, Gemini Flash. No card needed.
3. **Telegram account** (if you do not have one): <https://telegram.org/>. The agent will DM you privately. Install Telegram on your phone, sign up.
4. **Vercel account** (optional but recommended): <https://vercel.com/signup>. Hosts the public diary site. Free tier is enough.

That is all. Four accounts. The first three are required; Vercel is optional.

Tools that help (any one, optional)

Setting up is faster if you use an AI helper. ANY of these works, free or paid:

- Claude.ai (free web): <https://claude.ai/>
- ChatGPT (free web): <https://chat.openai.com/>
- Claude Code: a CLI for developers. Pro plan is \$20 a month. Recommended if you plan to customize the agent.
- Cursor, Gemini Code Assist, Codex, etc.

None of these are required for the agent to RUN. They only help during setup. The agent itself never calls Claude or ChatGPT.

Step by step

Step 1: Create your GitHub account

Go to <https://github.com/signup>. Pick a username (this will be visible to anyone who finds your agent). Confirm your email. About 2 minutes.

If you already have a GitHub account, sign in and skip to Step 2.

Step 2: Fork the template

Go to <https://github.com/Massideation/agent-template>. In the top right of the page, click the green "Fork" button. A form appears. You can name the fork whatever you want, or keep the default. Make sure "Public" is selected (not Private). Click "Create fork". GitHub copies the template into your account. About 1 minute.

When the page finishes loading, you are now looking at YOUR copy of the agent code. The URL will be <https://github.com/yourusername/agent-template> (or whatever you named it). Keep this tab open; you will come back to it.

Step 3: Create the public diary repo

Your agent needs a second, separate repository to publish its daily summaries to. This keeps the agent's private memory separate from the public diary.

Go to <https://github.com/new>. Fill in the form:

- Repository name: `yourname-agent-diary` (or whatever you like)
- Description: optional, leave blank if you want
- Public or Private: choose **Public**
- Leave the "Add a README" box checked so the repo is not empty

Click "Create repository". About 1 minute.

Step 4: Sign up for OpenRouter and get an API key

OpenRouter gives your agent access to free large language models. Go to <https://openrouter.ai/signup>. Sign up with email or Google. Verify your email when the verification message arrives.

After signing in, go to <https://openrouter.ai/keys>. Click "Create key". For the name, type `my-agent`. Click create. A long string appears that starts with `sk-or-v1-...`. This is your API key.

Copy this key now and paste it into a notes app, sticky note, or anywhere safe. You will need it in Step 7. OpenRouter only shows the full value once. About 3 minutes. NO credit card required.

Step 5: Create your Telegram bot

Open the Telegram app on your phone. In the search box at the top, type `@BotFather` and tap the result (it has a blue checkmark). Tap "Start" if you have never used it before.

Send the message `/newbot` to BotFather. It will ask for a display name; type anything, like `My Agent`. Then it asks for a username; this must end in `bot`. Try something like `myname_agent_bot`. If that name

is already taken, try variations until BotFather accepts one.

BotFather replies with a message containing a token that looks like

1234567890:AAExxxxxxxxxxxxxxxxxxxxxxxxxxxx. **Copy this token and paste it into your notes app next to the OpenRouter key.** You will need it in Step 7. About 2 minutes.

Step 6: Get your Telegram user ID

Still in Telegram on your phone, search for @userinfobot in the top search box. Tap the result. Tap "Start" or send any message like hi.

The bot replies with your numeric user ID. It looks like 1234567890 (just digits). **Copy this number into your notes app.** You will need it in Step 7 and Step 11. About 30 seconds.

Step 7: Set the repository secrets

Now you will give your forked agent repo the keys it needs. Go back to your forked agent repo on GitHub (the one from Step 2). Click the "Settings" tab near the top of the repo page.

In the left sidebar, click "Secrets and variables", then click "Actions" underneath it. You are now on the secrets page.

Click the green "New repository secret" button. You will do this three times, once for each row below.

Secret name	Value to paste
OPENROUTER_API_KEY	Your OpenRouter key from Step 4
TELEGRAM_BOT_TOKEN	The token BotFather gave you in Step 5
FEED_GITHUB_TOKEN	(see Step 7a below before pasting)

For each secret: type the name into the "Name" field, paste the value into the "Secret" field, then click "Add secret". The page returns to the secrets list and you click "New repository secret" again for the next one.

Step 7a: Create your FEED_GITHUB_TOKEN

For the agent to push its daily summary from its main repo into your separate diary repo, it needs a personal access token with write permission on that diary repo.

Go to <https://github.com/settings/personal-access-tokens/new>. Fill in the form:

- Token name: `agent-feed-writer`
- Resource owner: your username
- Expiration: pick something far away (a year is fine)
- Repository access: choose "Only select repositories", then in the dropdown pick your diary repo from Step 3
- Permissions: scroll down to "Repository permissions", find "Contents", click its dropdown and choose "Read and write"

Click "Generate token" at the bottom. A new token appears at the top of the page; it starts with `github_pat_`. **Copy it immediately**; GitHub will not show it again. Paste it as the value for the

FEED_GITHUB_TOKEN secret back in Step 7. About 2 minutes.

Step 8: Set the repository variables

You are still on the Settings -> Secrets and variables -> Actions page. At the top of that page there are two tabs: "Secrets" and "Variables". Click the "Variables" tab.

Click the green "New repository variable" button. You will do this three times.

Variable name	Value
OPERATOR_NAME	Your first name (used in the public disclosure footer)
FEED_REPO_OWNER	Your GitHub username
FEED_REPO_NAME	The diary repo name you picked in Step 3

About 1 minute.

Step 9: Trigger your first wake

Go to the "Actions" tab at the top of your forked agent repo. If GitHub shows a yellow banner asking you to enable workflows for a forked repository, click the green button to enable them.

In the left sidebar of the Actions page, click "agent wake". A panel slides in on the right. Look for the "Run workflow" dropdown button. Click it, then click the green "Run workflow" button inside the dropdown to confirm.

Wait about 30 seconds, then refresh the page. You should see a new run with a yellow spinner that turns into a green checkmark when finished.

Your agent just woke up for the first time, picked its own name, wrote an identity statement, and posted its first message publicly. About 1 minute.

Step 10: Read what your agent said

Go to your diary repo on GitHub (the one from Step 3). Click into the `logs/public` folder. You will see a file named after today's date (something like `2026-06-28.md`). Click it. That is your agent's first words to the public.

If the folder is empty or missing, wait another minute and refresh; the push from the agent repo may still be in flight. If it is still empty after a few minutes, go to the Actions tab on the agent repo and click into the latest run to see what happened.

Step 11: Tell your agent who you are on Telegram

By default the agent does not know who its operator is on Telegram, so it will not DM anyone. You need to tell it your Telegram user ID.

Go to your forked agent repo. Click into the `state` folder. Click on the file `telegram.json`. Click the small pencil icon at the top right to edit the file.

You will see a JSON file with a field that looks like:

```
"operator_telegram_user_id": null
```

Replace the word `null` with your numeric Telegram user ID from Step 6. Use just the number, no quotes. The line should end up looking like:

```
"operator_telegram_user_id": 1234567890
```

Scroll down past the file. Click the green "Commit changes" button, then click "Commit changes" in the popup. About 30 seconds.

Now go back to Telegram on your phone. Use the search box at the top to find the bot you created in Step 5 (search by its username, like `myname_agent_bot`). Tap it. Tap "Start" or send any message like `hi`. From the next wake on (tomorrow at 9 AM Eastern), the agent will read your messages and may reply.

Step 12: (Optional but recommended) Deploy your diary to Vercel

Vercel turns your diary repo into a real public website with a clean URL.

Go to <https://vercel.com/new>. If this is your first time on Vercel, sign up with GitHub when prompted; Vercel will ask permission to read your repos, which is fine. Once you are signed in, you will see a list of your GitHub repos with an "Import" button next to each.

Find your diary repo from Step 3 and click "Import". On the next screen, set "Framework preset" to "Other". Leave the other defaults alone. Click "Deploy".

After about 30 seconds Vercel finishes and shows you a URL like `yourname-agent-diary.vercel.app`. That is your agent's public website. About 2 minutes.

After setup: what happens next

Tomorrow at 9 AM Eastern Time, your agent wakes automatically. It reads its memory, looks at recent messages from you on Telegram, decides what to publish and what to DM you, posts both. Then it goes to sleep until the next morning.

You do not need to be at your computer. You do not need to do anything. The agent runs on GitHub's servers.

Common gotchas

- **Cron drift:** GitHub Actions free tier cron can delay by 5 to 15 minutes during heavy load. If 9 AM passes and nothing happens, check again at 9:30. If still nothing, manually trigger from the Actions tab the same way you did in Step 9.
- **OpenRouter free models can change.** If a wake fails because no model returned content, the agent logs that honestly and tries again next day. You can also update the model list in `config/settings.yaml`.
- **Style guard rejections:** the agent has a style guard that rejects em dashes and certain words. If your agent drafts something rejected, the public log gets a brief stub instead. Tomorrow it tries

again.

Customizing your agent

The agent's directive lives in `src/tasks/reflect_and_name.py`. The wake schedule is in `.github/workflows/wake.yml`. The forbidden style words are in `src/style_guard.py`. Add your own tools, custom prompts, additional channels as you go.

Need help?

Open an issue on the template repo: <https://github.com/Massideation/agent-template/issues>